

Sushant Twayana

AI/ML Engineer, MLOps, Data Science

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Proven expertise in deploying enterprise applications using LangChain, LangGraph, LangSmith RAG, Kubernetes, and AWS infrastructure with proven track record of optimizing model performance and reducing latency.

PROFESSIONAL EXPERIENCE

Junior AI/ML Engineer | VANILLA TRANSTECHOR (July 2025 - Present)

- Architected end-to-end multilingual speech processing pipeline integrating Whisper transcription, custom translation model & Rasa intent classification, improving **ASR** accuracy & reducing latency
- **Fine-tuned Whisper** models using **LoRA adaptation** for domain-specific optimization, achieving 25% performance improvement over baseline AWS Transcribe
- Working on research and production of **OCR for documents** in fintech domain.
- Collaboratively implemented real-time **fraud detection algorithms** for fintech transaction data using collaborative ML approaches.

Associate Data Scientist | Dlytica – Data Analytics and AI (Dec 2024 - May 2025)

- Deployed enterprise **MLOps pipelines** using Kubernetes and Docker with automated CI/CD for model versioning and deployment on **KubeFlow**.
- Built **production conversational AI systems** leveraging LangChain, RAG architecture, and autonomous agent frameworks for banking clients.
- Engineered real-time **banking recommendation system** and **customer segmentation solution**. processing large-scale data with **PySpark** and **Apache Superset**.

Training Assistant: Served as Training Assistant for Data Science programs, mentoring 50+ aspiring data scientists and contributing to curriculum development

AI/ML Engineer Intern | Comfort Yantra (July 2024 - October 2024)

- Developed **pre-production-ready conversational AI** using RAG architecture with LangChain, FastAPI, GPT-3.5 Turbo, and Chroma vector store.
- Built hybrid automation solutions combining LLM-powered and regex-based methods for e-commerce product categorization and **NL-to-SQL translation**.
- Integrated Azure OpenAI and Meta LLaMA-3-8B for semantic parsing and multitool agent.

TECHNICAL SKILLS

- **LLM & GenAI:** LangChain, LangGraph, RAG, OpenAI GPT, Ollama, Vector Databases (Chroma, FAISS), MCP, LangSmith, CrewAI
- **MLOps & Cloud:** Kubernetes, Docker, Kubeflow, KServe, AWS EC2, CI/CD Pipelines
- **ML/DL Frameworks:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, PySpark
- **Specialized Models:** Whisper, Rasa, YOLO, BERT, T5, BLIP, GANs
- **Development:** Python, FastAPI, gRPC, SQL, Git

PROJECTS

Agentic AI & Multi-Agent Systems

Production-Grade Agentic Chatbot with LangGraph | [GitHub](#)

- GPT-like conversational system with LangGraph orchestration, streaming, persistent threading, RAG document upload, human-in-the-loop, and LangSmith observability.

Multi-Agent Blog Generation System | [GitHub](#)

- Autonomous workflow with conditional routing, state management, tool calling, and custom graph nodes for multilingual content generation.

GenAI & LLM Applications

Enterprise RAG System & Intelligent Chatbots | LangChain: [GitHub](#) | [git](#)

- Production-grade RAG pipelines with OpenAI/Ollama integration, LCEL, vector databases (Chroma/FAISS), conversation memory, and advanced document processing.

AI-Powered Expense Tracking Application | (MCP) Integration: [GitHub](#)

- Financial management app using Model Context Protocol for standardized AI-system communication with automated budget monitoring.

Newari Food Assistant Chatbot (Dialogflow): [GitHub](#)

- Architected agents and integrated dialogflow for natural conversation and intent recognition

MLOps & Cloud Deployment

Scalable MNIST Classification Pipeline (Kubeflow + KServe): [GitHub](#)

- Production ML pipeline with KubeFlow orchestration, KServe inference, and automated CI/CD for model versioning.

Fixed Deposit Recommendation System & Customer Segmentation | Banking Domain

- PySpark-based system with ML hyperparameter optimization and Apache Superset for distributed processing and visualization.

Computer Vision & Deep Learning & NLP

Text-to-Face Generation System (DC GAN & DF GAN): [\(GitHub\)](#)

- Implemented generative AI system using for human face generation from text descriptions.

Food Recognition & Recipe Generation System | ResNet50, YOLOv8, BLIP, T5: [\(GitHub\)](#) | [\(link\)](#)

- End-to-end pipeline with ResNet50/YOLOv8 detection, BLIP visual understanding, and T5-based recipe generation.

Content-Based Movie Recommender: [\(GitHub\)](#)

- NLP-based recommendation system using embeddings and cosine similarity algorithms.

Python And FastAPI Development

Distributed Restaurant Order System (FastAPI + gRPC): [\(GitHub\)](#)

- Microservices architecture with FastAPI gateway, gRPC communication, Protocol Buffers, and containerized deployment.

Insurance Premium Prediction API: [\(GitHub\)](#)

- FastAPI ML inference service with Pydantic validation, Streamlit frontend, and Random Forest model.

ACHIEVEMENTS

- **Winner:** ProtoBytes Hackathon at ACEM 2024: Developed innovative AI solution under competitive deadline.
- **Best Presenter Award** - KU IT FEST 2024
- **Event Organizer** - Hult Prize Event 2022/23: Led team in executing large-scale event.
- **Associate** - ITSNP: Active contributor to technical and community-building initiatives (2022).

EDUCATION

Bachelor's in computer engineering, Khwopa Engineering College | (Oct 2019 – Aug 2024)

CGPA: 3.50/4.0

CERTIFICATIONS

- **Complete Agentic AI Bootcamp With LangGraph and Langchain** (KRISHAI Technologies)
- **Associate Data Scientist in Python Certification** (DataCamp)
- **PyTorch and Keras (TensorFlow) Full Course**
- **Advanced Python and Data Science Certifications** (DataCamp)